

The simplicity of Duplicity

Although servers rely entirely on the command line, desktop users no longer do. But there is still a small community of people who like to explore the power and beauty of it. So let us get to explore Duplicity, the command line backend that powers Déjà Dup.

Duplicity supports GnuPG encryption and allows us to use local or remote storage methods (such as FTP, SFTP, SCP, WebDAV, WebDAVs, Google Docs, HSI and Amazon S3). It is powered by *rsync*, a fast and reliable file copying tool, the algorithm for which helps in efficient data transfer. Duplicity supports many features including deleted files, full UNIX permissions and symbolic links, excluding hard links.

And the good news is that it is not really hard to use.

Warning

Duplicity may crash while copying */proc* if you are backing up the root directory */*. Use *--exclude=/proc* to avoid this.

Installation

Open a terminal, type *duplicity* and press the *Enter* key to see something (probably an error message) so that you can ensure it is installed in your system. If not, use a package manager or give the following commands:

```
sudo apt-get install duplicity
sudo apt-get install ncftp
```

The second package is to get FTP support. Yes, it is optional.

Using Duplicity

The following example illustrates how we can store the backup of */home/nandakumar/test* in */home/nandakumar/backup*. Remember that the latter should be replaced with some external storage device or there is no use of taking a backup. We won't go into that right now for the sake of simplicity.

Here is the command to be used for this purpose:

```
duplicity test file:///home/nandakumar/backup
```

We skipped */home/nandakumar* from the path of *test* since the terminal is in the home folder, by default (otherwise use a full path or *cd*). However, Duplicity insists that one of the paths must be a URI. That's why we left the second one intact.

Once this command is executed, the program asks for an encryption passphrase. Give something powerful and remember it.

If this command is run repeatedly, Duplicity copies changes since the last backup only. If you want to take a full backup, use the option *full*, as follows:

```
duplicity full test file:///home/nandakumar/backup
```

Restoration can also be done simply, using the following command:

```
duplicity file:///home/nandakumar/backup test
```

If you'd like to exclude some files/sub-folders from being backed up, use the *--exclude* option, as follows:

```
duplicity --exclude subfolder1 test file:///home/nandakumar/backup
```

The non-interactive route

From the above examples, we've seen that most of the time when we run Duplicity, it asks for passwords. This is inconvenient if we are writing a shell script. There should not be any prompt at all. That's why Duplicity supports inputs via environment variables. Let's look at the following example:

```
export PASSPHRASE=MyEncryptionPassword
export FTP_PASSWORD=MyFTPPassword
duplicity test ftp://username@ftp.domain.com/backup
unset PASSPHRASE
unset FTP_PASSWORD
```

In the above example, the folder *test* is backed up into another folder *backup*, which resides in *ftp.domain.com*. If you use a local storage, there is no need to set an *FTP_PASSWORD*.



Note: These lines can appear inside your bash history file. Delete them for security

A final word

We now know the basics of two backup utilities. However, I suggest you read up on Duplicity to find out more about its hidden power (run the command *man duplicity*).

If you work on some long-term projects, make use of these tools. But it is better to stick to the classic copy-paste method once you have the final product (if you have completed writing a novel, for instance). This helps to avoid software dependencies. 

References

- <https://wiki.gnome.org/Apps/DéjàDup>
- <http://duplicity.nongnu.org/duplicity.1.html>
- <https://help.ubuntu.com/community/DuplicityBackupHowto>

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